

Solving Inequalities in Real-World ContextsAdditional Practice 7A.01.10 **ANSWER KEY**

Jaxon solved the inequality $32 < -4(-1.4n - 22)$ and made more than one error. His work is shown. Use it for questions 1 through 3.

Jaxon's work

$$\begin{aligned}
 32 &< -4(-1.4n - 22) \\
 32 \div (-4) &< (-4 \div -4)(-1.4n \div -4 - 22 \div -4) \\
 -8 &< 1(0.35n + 5.5) \\
 -8 &< 0.35n + 5.5 \\
 -13.5 &< 0.35n \\
 -270/7 &< n
 \end{aligned}$$

1. What errors did Jaxon make? There is more than one.

First: he divided both sides by -4 and did not reverse the symbol. Dividing by a negative always reverses it.

Second: he divided the right side by -4 twice — once on the factor out front and again on each term inside the parentheses. Dividing $4(x)$ by 4 gives x , not $x/4$.

2. Write a brief note to Jaxon explaining how to avoid this in the future. Say what he did correctly as well.

Sample: "Choosing to divide by -4 instead of distributing was a good call — it is the shorter route here. Two things to watch. Dividing by a negative reverses the symbol, so that first line should read $-8 > \dots$. And when you divide the right side by -4 , the -4 out front disappears; you do not also divide the terms inside. Your arithmetic on $22 \div 4$ and $1.4 \div 4$ was correct, which is why the mistake is hard to see."

Answers will vary. A complete note names both errors, gives a working alternative and identifies something correct.

3. Solve Jaxon's inequality correctly and represent the solution on the number line.

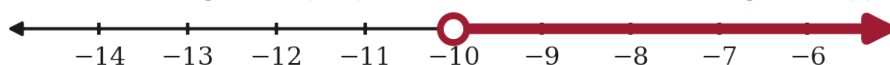
$$32 < -4(-1.4n - 22)$$

$$-8 > -1.4n - 22 \quad (\text{divide both sides by } -4 \text{ and REVERSE})$$

$$14 > -1.4n \quad (\text{add } 22 \text{ to both sides})$$

$$-10 < n, \text{ so } n > -10 \quad (\text{divide both sides by } -1.4 \text{ and REVERSE again})$$

$$\text{Check: } n = 0 \text{ gives } -4(-22) = 88 \text{ and } 32 < 88 \checkmark; \quad n = -20 \text{ gives } -4(6) = -24 \text{ and } 32 < -24 \times$$



Open circle at -10 , shaded right. Distributing instead gives $32 < 5.6n + 88$, which lands in the same place with no reversal at all — every step there divides by a positive number, and turning $-10 < n$ round into $n > -10$ is a rewrite, not a reversal.

For questions 4 through 7, solve the inequality and represent the solution on the number line.

4. $(5/3)(-a + 3/4) \leq -2$

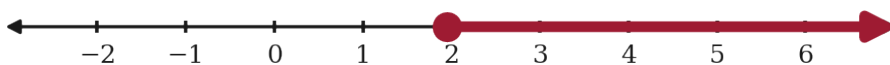
$-a + 3/4 \leq -1.2$ (multiply both sides by $3/5$ — no reversal)

$-a \leq -1.95$

$a \geq 1.95$ (multiply both sides by -1 and REVERSE)

Check: $a = 3$ gives $(5/3)(-2.25) = -3.75$ and $-3.75 \leq -2$ ✓; $a = 0$ gives $(5/3)(0.75) = 1.25$ and $1.25 \leq -2$ ✗

Solution. $a \geq 1.95$



The boundary sits just left of the 2 tick mark. Closed circle a hair before 2, shaded right — and say “just below 2” in words.

5. $-3(2t - 4/3) > 6$

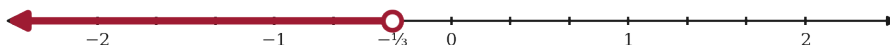
$2t - 4/3 < -2$ (divide both sides by -3 and REVERSE)

$2t < -2/3$

$t < -1/3$

Check: $t = -1$ gives $-3(-2 - 1.333) = 10$ and $10 > 6$ ✓; $t = 0$ gives $-3(-1.333) = 4$ and $4 > 6$ ✗

Solution. $t < -1/3$



The scale on this line runs in thirds so the boundary lands on a tick. Open circle at $-1/3$, shade left.

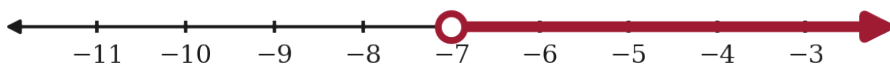
6. $(-6.2 + m)/4 > -3.3$

$-6.2 + m > -13.2$ (multiply both sides by 4 — no reversal)

$m > -7$

Check: $m = 0$ gives -1.55 and $-1.55 > -3.3$ ✓; $m = -10$ gives -4.05 and $-4.05 > -3.3$ ✗

Solution. $m > -7$



The whole numerator is divided by 4, so multiply the whole other side by 4 before doing anything else.

7. $(2/5)k - 8.4 \leq -4$

$(2/5)k \leq 4.4$ (add 8.4 to both sides)

$k \leq 11$ (multiply both sides by $5/2$)

Check: $k = 11$ gives $4.4 - 8.4 = -4$ and $-4 \leq -4$ ✓; $k = 15$ gives $6 - 8.4 = -2.4$ and $-2.4 \leq -4$ ✗

Solution. $k \leq 11$



$2/5 = 0.4$ if you would rather use a decimal. Either route is fine and neither reverses.

When a store had sold $\frac{3}{5}$ of the shirts on display, it brought out another 40 from the stock room. The store likes to keep at least 170 shirts on display. The manager wrote $(\frac{2}{5})x + 40 \geq 170$, where x is the number of shirts that were originally on display.

8. Explain what the $\frac{2}{5}$ means in this inequality.

Three fifths of the original shirts were sold, so two fifths of them are still on display.

$(\frac{2}{5})x$ is the number of the original x shirts that remain. The 40 brought out from the stock room is added to that.

9. Solve the inequality.

$$(\frac{2}{5})x \geq 130 \quad (\text{subtract 40 from both sides})$$

$$x \geq 325 \quad (\text{multiply both sides by } \frac{5}{2})$$

$$\text{Check: } x = 325 \text{ gives } (\frac{2}{5})(325) + 40 = 130 + 40 = 170, \text{ and } 170 \geq 170 \checkmark$$

10. Explain what the solution means in terms of the situation.

The store must have started with at least 325 shirts on display.

With fewer than 325 to begin with, two fifths of them plus the 40 from the stock room would not reach 170.

325 is a minimum, so round UP if a decimal ever appears here. Shirts are whole objects and 325 is already whole, so nothing needs rounding this time.